

Interpretable Machine learning for ECG Signal Quality Classification: A comparative Analysis of
Logistic and Deep Learning Approaches

Final Project Report

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Abstract

Electrocardiogram (ECG) signal quality assessment is an essential component of automated cardiac diagnostic systems, yet it receives very little attention. Poor signal quality arises from various artifacts, including baseline drift, electrode motion artifacts, and muscle noise, which can compromise the reliability of a clinical algorithm, leading to misdiagnoses or missed diagnoses. This project investigates whether interpretable, literature-based signal-quality metrics can reliably discriminate between clean and degraded ECG segments under realistic clinical noise conditions.

Using the PTB-XL dataset and expert noise annotations, five classification models, ranging from interpretable logistic regression to state-of-the-art deep learning architectures, were developed and benchmarked. Two interpretable logistic models (Models A and B) employed L1-regularized logistic regression on handcrafted signal quality indices (SQIs), achieving ROC-AUC scores of 0.559 and 0.556, respectively. This was followed by deep learning models, which progressively improved performance. A baseline multilayer perceptron (DL1) achieved 0.555 AUC, a deep 1D-CNN operating on raw waveforms (DL2) reached 0.797 AUC, and the ultimate benchmark, a CNN-Transformer ensemble with uncertainty quantification (DL3), achieved the highest performance at 0.813 AUC.

Findings revealed a fundamental limitation of handcrafted SQI features for ECG quality assessment, with performance plateauing at around 56% AUC across classification algorithms. In contrast, end-to-end deep learning models trained on raw 12-lead waveform data captured signal degradation patterns that feature engineering could not detect. The DL3 ensemble additionally provides calibrated uncertainty estimates critical for clinical deployment, where knowing when the model is uncertain is as important as its predictions. These results suggest that future ECG quality assessment systems should leverage deep learning architectures while incorporating uncertainty quantification to support trustworthy clinical decision-making.

Background

Cardiovascular diseases are among the leading causes of mortality and disability worldwide (Di Cesare et al., 2024). Early diagnosis may lower the risk of diseases such as acute myocardial infarction and subsequent death in people with myocardial infarction (Almansouri et al., 2024). One of the tools for early detection is Electrocardiography (ECG), often combined with other diagnostic methods (Almansouri et al., 2024; Goldenthal et al., 2019). ECG is a critical non-invasive tool for cardiovascular diagnosis (Strodthoff et al., 2021). ECGs guide a range of clinical decisions, from routine screening all the way to emergency care, and influence the clinical care of millions of patients. The standard twelve-lead ECG records the heart's electrical activity as vectors from different orientations (Kligfield et al., 2007), enabling detection of arrhythmias, ischemia, conduction abnormalities, and structural heart disease. As healthcare systems increasingly rely on automated ECG interpretation algorithms, from simple heart rate detection to complex arrhythmia classification, the quality of the input signal becomes very important (Strodthoff et al., 2021; Wagner et al., 2020).



Figure 1. Representative examples of clean and degraded 12-lead ECG waveforms from the PTB-XL dataset, illustrating common signal quality artifacts including baseline drift, static noise, and electrode problems.

Clinical ECG recordings obtained in the real world are hardly free from distortions. In routine care, signals are frequently affected by different sources of degradation, which include baseline drift caused by patient respiration and electrode impedance changes, muscle artifacts from patient movement or tremor, powerline interference from electrical equipment, and electrode contact problems from poor skin preparation or patient motion (Wagner et al., 2020). Previous studies have shown that a substantial proportion of recorded clinical ECGs contain significant noise artifacts that can compromise diagnostic accuracy (Orphanidou et al., 2015; Satija et al., 2018). In the PTB-XL dataset used in this study, approximately 23% of recordings were labelled by cardiologists as exhibiting some form of signal degradation. This problem is even more pronounced in ambulatory systems, intensive care units, and wearable device contexts where

motion artifacts are prevalent (Iqbal et al., 2021). The consequences of poor signal quality go beyond just inconvenience. Degraded ECGs can conceal subtle ST-segment changes indicative of myocardial ischemia, mask atrial fibrillation episodes, or trigger false alarms that add to alert fatigue in a clinical environment (Clifford et al., 2012; Drew et al., 2014). Algorithms trained on clean data often produce unreliable outputs when confronted with degraded signals, potentially leading to both false positives that waste clinical resources and false negatives that delay necessary interventions.

Motivation and Scope

This project aims to address the key question: Can we distinguish between clean and degraded ECG recordings accurately while ensuring trustworthiness for clinical deployment? This question was addressed by systematically comparing interpretable machine learning methods (L1-regularized logistic regression with handcrafted features) with deep learning approaches (ranging from simple neural networks to state-of-the-art transformer architectures). Aside from classification, a key metric was to quantify uncertainty (i.e., knowing when the model is confident versus uncertain), which helps achieve the second part of the project aim and is important for trustworthiness in clinical decision support.

Research Question

Our primary research question is: Can interpretable, literature-based signal-quality metrics reliably discriminate between clean and degraded ECG signals under real-world clinical noise conditions? More specifically, we divide the research questions into 4 unique parts:

1. How well do traditional signal quality indices (SQIs) derived from the ECG signal quality literature perform when combined with interpretable classification algorithms?
2. To what extent can deep learning models operating on raw waveform data outperform handcrafted feature approaches?
3. Can we quantify model uncertainty for clinical decision-making?
4. What architectural choices (CNN, Transformer, ensemble) yield the best performance-interpretability trade-off?

Current approaches to ECG signal quality assessment are not without their shortcomings, and that motivated this work. Traditional SQI-based approaches (Li et al., 2008; Zhao & Zhang, 2018) often depend on manually selected thresholds that fail to generalize across recording environments, patient groups, and acquisition systems. Based on the aforementioned reasons, a threshold optimized for one dataset often performs poorly on another. Additionally, many traditional methods assign a single quality label to entire ECG recordings, overlooking transient artifacts that may affect only specific leads or time segments, thereby conflating intermittent

noise with sustained signal degradation. Furthermore, deep learning models for ECG are typically designed for diagnostic tasks, such as arrhythmia detection and myocardial infarction identification, rather than for signal quality assessment. The features learned for diagnosis may not capture the noise patterns relevant to quality assessment, as representations happen to be task-specific (Strodthoff et al., 2021). Finally, most existing systems provide only point estimates without accompanying confidence estimates. In clinical settings, a model with uncertainty estimates is crucial for appropriate human oversight (Rahman et al., 2022).

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Data and Experimental Setup

The PTB-XL dataset (Wagner et al., 2020), the largest publicly available clinical 12-lead ECG dataset, was used to answer the research questions. The dataset comprised 21,837 records from 18,885 patients collected at the Physikalisch-Technische Bundesanstalt in Germany. Each recording is 10 seconds in duration, sampled at 500 Hz, providing 5,000 samples per lead across all 12 standard leads (I, II, III, aVR, aVL, aVF, V1-V6). All recordings were labelled by expert cardiologists for signal quality issues, including baseline drift, static noise, burst noise, and electrode problems.

Binary quality labels were generated by aggregating the noise annotations. Recordings containing any non-zero annotation for baseline drift, static noise, burst noise, or electrode problems were labeled as degraded. In contrast, records with all-zero noise annotations were labeled as clean. This resulted in a total of 16,816 clean recordings (77.0%) and 5,021 degraded recordings (23.0%), creating a moderately imbalanced classification task that reflects realistic clinical distributions.

Data splitting

The PTB-XL dataset provides predefined or proposed stratified folds that maintain comparable distributions across splits while preserving patient boundaries, ensuring that no patient appears in both the training and test sets. In this study, the recommended split was adopted with folds 1-8 for training (17418 recordings), fold 9 for validation (2203 recordings), and fold 10 for final testing (2216 recordings). Of the 2216 recordings in fold 10, 2203 were successfully processed across all pipelines and used for the final evaluation. The validation set was used for hyperparameter tuning, threshold optimization, and model selection. In contrast, the test set was held out for final performance evaluation to prevent overfitting to the evaluation data.

Evaluation metrics

Given the class imbalance and clinical relevance of misclassification, model performance was assessed using a comprehensive set of metrics beyond overall accuracy. The area under the receiver operating characteristic curve (ROC-AUC) was used as the primary measure of discrimination to estimate the discrimination ability across all possible classification thresholds. The area under the precision-recall curve (AUPRC) was also reported to emphasize the minority class (degraded recordings). Sensitivity (Recall) was also used to estimate the proportion of degraded recordings correctly identified, which is important for detecting signal quality issues. Specificity on the other hand, quantified the proportion of clean recordings correctly identified, it was necessary to avoid unnecessary reacquisition. Expected Calibration Error (ECE) was used to estimate how well predicted probabilities match observed frequencies, with lower values

indicating better calibration. For uncertainty-quantified models, the proportion of test samples whose true labels fall within the prediction set was assessed with conformal coverage.

Threshold optimization

Instead of using the default 0.5 classification threshold, a systematic threshold optimization was applied to the validation set. Three strategies were examined (1) optimization based on the Youden's J statistic (maximizing sensitivity + specificity - 1), which was adopted as the primary approach; (2) F2-score optimization, which weights recall higher than precision; and (3) cost-sensitive thresholding, assuming false negatives corresponding to missed degraded signal, incurring a penalty five times greater than that of false positives. All performance on the test metrics was reported using the Youden-optimal threshold determined on validation data.

Signal preprocessing

Before feature extraction, all ECG signals underwent standardized clinical preprocessing. Baseline drift was corrected using median filtering with a 200-ms window, followed by a 60 Hz notch filtering ($Q=30$) to suppress powerline interference. A low-pass Butterworth filter (cutoff 40 Hz, order 3) attenuated high-frequency muscle noise while preserving relevant ECG morphology. This preprocessing pipeline was applied consistently across all models to ensure fair comparison.

Models

Five models of increasing complexity were developed, ranging from an interpretable feature-based approach to sophisticated deep learning models. This progression enables the assessment of the incremental contribution related to each methodological improvement

Model A: Basic SQI features

The first interpretable model employed 15 carefully selected signal quality indices drawn from the ECG signal processing literature (Behar et al., 2013; Li et al., 2008; Zhao & Zhang, 2018). These features capture different aspects of signal quality. Statistical features such as the signal mean, standard deviation, root-mean-square amplitude, peak-to-peak range, skewness (sSQI), and kurtosis (kSQI). For instance, the kurtosis of a clean ECG tends to be higher due to the sharp QRS complex, whereas noisy signals have lower kurtosis, approaching that of a Gaussian distribution. Hjorth parameters, such as mobility and complexity metrics, captured spectral characteristics and signal irregularity, respectively. Spectral features comprised total signal

power, power in the QRS frequency band (5-40 Hz), baseline power (0-1 Hz), and high-frequency noise power (40-100 Hz). The perturbation-based Signal Quality Index (pSQI) index is defined as the ratio of QRS-band power to broadband power (0-40Hz), which indicates whether the signal energy is concentrated in the physiologically relevant frequency range. The Baseline Power Signal Quality Index (basSQI) index measures baseline stability by quantifying low-frequency contamination. All features were extracted per-lead, then aggregated across leads using mean and standard deviation, and further aggregated across limb leads (I, II, III, aVR, aVL, aVF) and precordial leads (V1-V6) separately. After variance-based filtering and mutual information-based feature selection, the top 15 features were retained for classification. Classification was conducted using L1-regularized logistic regression ($C=0.1$) with balanced class weights to address the class imbalance. The L1 penalty was to induce sparsity, effectively performing feature selection and producing an interpretable model where each non-zero coefficient indicates feature importance.

Model B: Enhanced SQI features

The second model extended the basic feature set by incorporating eight additional indices intended to capture more subtle quality issues. The iSQI (inter-algorithm agreement SQI) which measures consistency between two different R-peak detection algorithms. A higher inconsistent score indicates that the signal may be too noisy for reliable beat detection. Heart rate variability features (i.e., mean RR interval, RR standard deviation, RMSSD, and coefficient of variation) capture physiological plausibility. Abnormally irregular RR intervals may indicate noise contamination rather than genuine arrhythmia. Spectral entropy quantifies the flatness of the power spectrum with higher entropy indicating a noisier signal. Inter-lead correlation features were computed by constructing a Pearson correlation matrix across all 12 leads and extracting summary statistics (mean and minimum correlation). Clean ECGs exhibit predictable inter-lead relationships based on the underlying cardiac electrical field; noise disrupts these correlations. The 20 most informative features were retained using the same mutual information procedure as Model A.

DL1: Simple Multilayer Perceptron

The first deep learning model was implemented as a baseline feature-based neural network. Using the same basic SQI features as Model A, a simple feedforward network was trained with the following the architecture: an input layer of size 15, followed by a dense layer with 128 units and ReLU activation, a dropout layer with rate 0.3, a dense layer with 64 units and ReLU activation, a second dropout layer with rate 0.2, and a final single-unit output layer with BCEWithLogitsLoss. The network was trained for 40 epochs using the Adam optimizer (learning rate of 0.001) with binary cross-entropy loss weighted by the inverse class frequency to account

for imbalance. This architecture tests whether non-linear transformations of the handcrafted features can improve upon logistic regression.

DL2: Deep 1D Convolution Neural Network

Moving beyond handcrafted features, DL2 operates directly on the raw 12-lead ECG waveform with an input dimension of 12 x 5000 samples. The architecture consisted of four convolutional blocks with progressively expanding channel dimensions (12->32->64->128->256), each containing a one-dimensional convolutional layer with a kernel size of 15 and stride of 2, followed by batch normalization, ReLU activation, max pooling with a factor of 2, and dropout with rates between 0.2 and 0.3. The fourth block, however, used a kernel size of 7 and a stride of 1 with adaptive average pooling and no dropout. This structure learns the discriminative representations directly from raw waveform data, eliminating the need for manual feature engineering. After feature extraction, adaptive average pooling was applied to reduce the temporal dimension to 1, producing a 256-dimensional feature vector. A two-layer classifier (256 -> 64 -> 1 with ReLU and dropout) makes the final prediction. The model was trained for 50 epochs with the AdamW optimizer (learning rate=0.001, weight decay=0.01), weighted BCE loss, and early stopping based on validation AUC, with a patience of 10 epochs.

DL3: CNN-transformer Ensemble

The final benchmark model combines several state-of-the-art techniques to achieve the best performance with calibrated uncertainty estimates. The front-end consisted of a multi-scale convolution that captures features at different temporal resolutions using parallel convolutions with kernel sizes 3, 7, 15, and 31. This inception-style module (inspired by Szegedy et al., 2015) captures both fine-scale QRS morphology and long-range rhythm patterns. Squeeze-and-excitation blocks adaptively recalibrate channel-wise feature responses. Two residual blocks with skip connections (He et al., 2016) enable training of deeper networks. Features extracted by the convolutional stem are treated as a temporal sequence and processed by a 4-layer Transformer encoder (Vaswani et al., 2017) with 8 attention heads and 128-dimensional hidden states. Self-attention enables the model to capture long-range dependencies across the 10-second recording and learn which temporal segments are most informative for quality assessment. Positional encoding preserves temporal ordering information. Instead of global average pooling, an attention-based pooling layer determines the contribution of each temporal position to the final representation. This allows the model to focus on the most informative segments while reducing the influence of uninformative regions. The final model comprises five independently trained models with different random initializations, forming an ensemble. Each model includes an uncertainty head that predicts data uncertainty alongside the classification output. Model uncertainty is estimated from the variability between ensemble predictions. This decomposition

distinguishes between uncertainty arising from inherently ambiguous inputs and that arising from a lack of training data. To address class imbalance, focal loss with $\gamma=2.0$ was employed, down-weighting well-classified examples and focusing learning on complex cases (Lin et al., 2017).

Post-hoc calibration was performed using temperature scaling (Guo et al., 2017), which adjusts predicted probabilities to better match empirical frequencies, reducing miscalibration without affecting discrimination. Split conformal prediction (Vovk et al., 2023) was applied to construct prediction sets with guaranteed marginal coverage at the $\alpha=0.10$ level. This provides distribution-free uncertainty quantification with finite-sample guarantees. The model was trained for up to 100 epochs per ensemble member using the AdamW optimizer (learning rate=0.0001, weight decay=0.01) with cosine annealing warm restarts. To enhance generalization, data augmentation methods such as random temporal shifts (± 500 samples), Gaussian noise injection ($\sigma=0.01$), amplitude scaling (0.9-1.1x), and random lead dropout (15% probability) were employed.

Results

Table 1 presents the test set results for all five models.

Model	ROC-AUC	AUPRC	Sensitivity	Specificity	PPV	NPV	ECE
Model A (Basic)	0.559	0.275	0.509	0.596	0.284	0.793	0.260
Model B (Enhanced)	0.556	0.276	0.561	0.516	0.268	0.788	0.259
DL1 (MLP)	0.555	0.289	0.577	0.505	0.269	0.791	0.312
DL2 (1D-CNN)	0.797	0.543	0.777	0.677	0.432	0.906	0.236
DL3 (Ensemble)	0.813	0.569	0.758	0.722	0.463	0.904	0.251

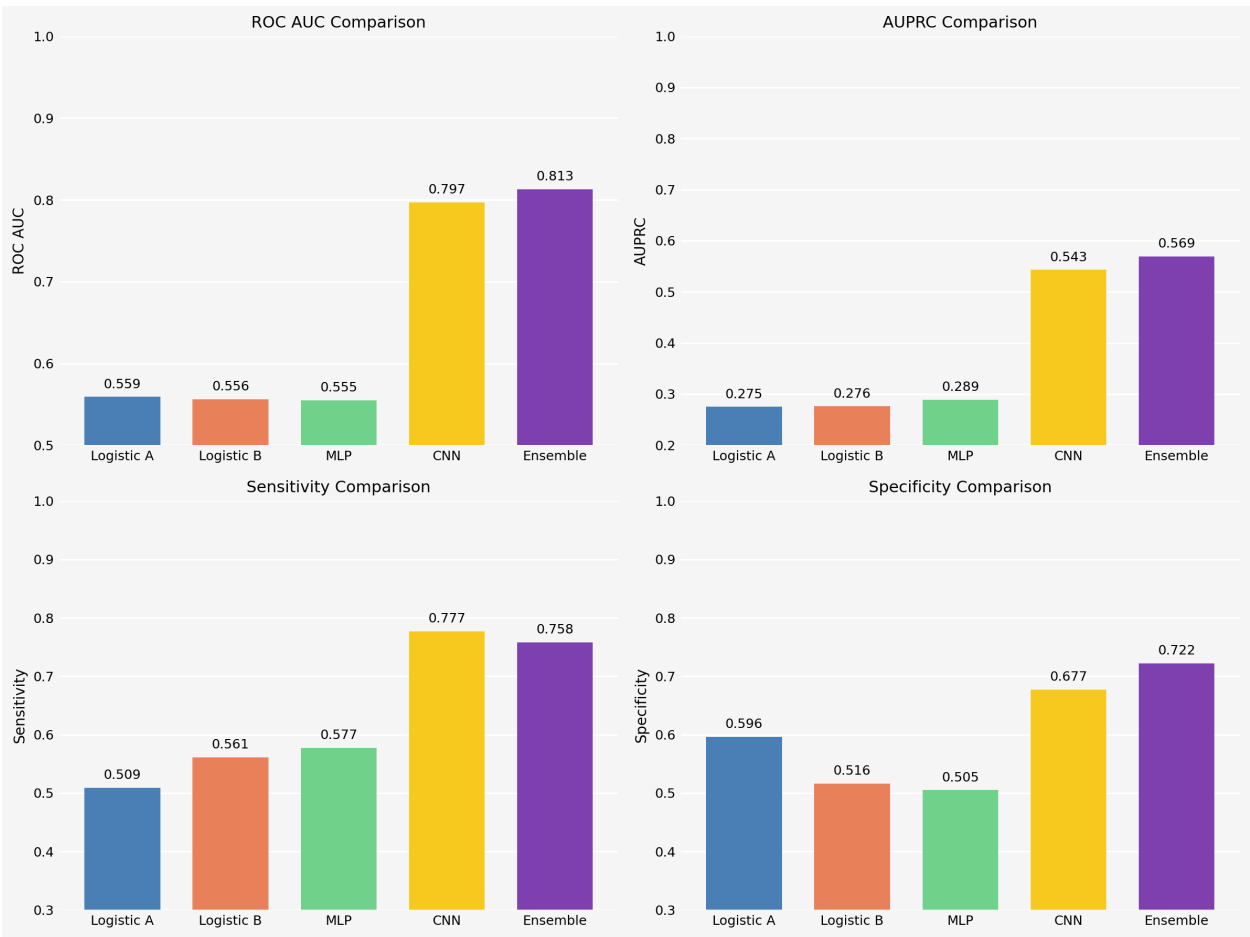


Figure 2. Final benchmark comparison across all five models on ROC-AUC, AUPRC, Sensitivity, and Specificity.

Feature engineering ceiling

The most interesting finding is the ceiling effect observed with handcrafted SQI features. Models A and B, along with DL1, all achieved nearly identical ROC-AUC scores (approximately 0.56) despite using different feature sets and classification algorithms. Adding the enhanced features to Model B had no effect on model performance relative to the basic features in Model A. Similarly, replacing logistic regression with a neural network (DL1) did not improve performance. This suggests that the information content of the handcrafted SQI features, rather than the classification algorithm, is the limiting factor.

This observation has important implications for automated ECG signal quality assessment. There has been a shift in focus to developing more sophisticated SQI combinations and the use of more complex classifiers on some of the existing features. Results from the current experiment suggest that fundamental progress requires moving beyond handcrafted features entirely, toward representations learned directly from raw waveform data.

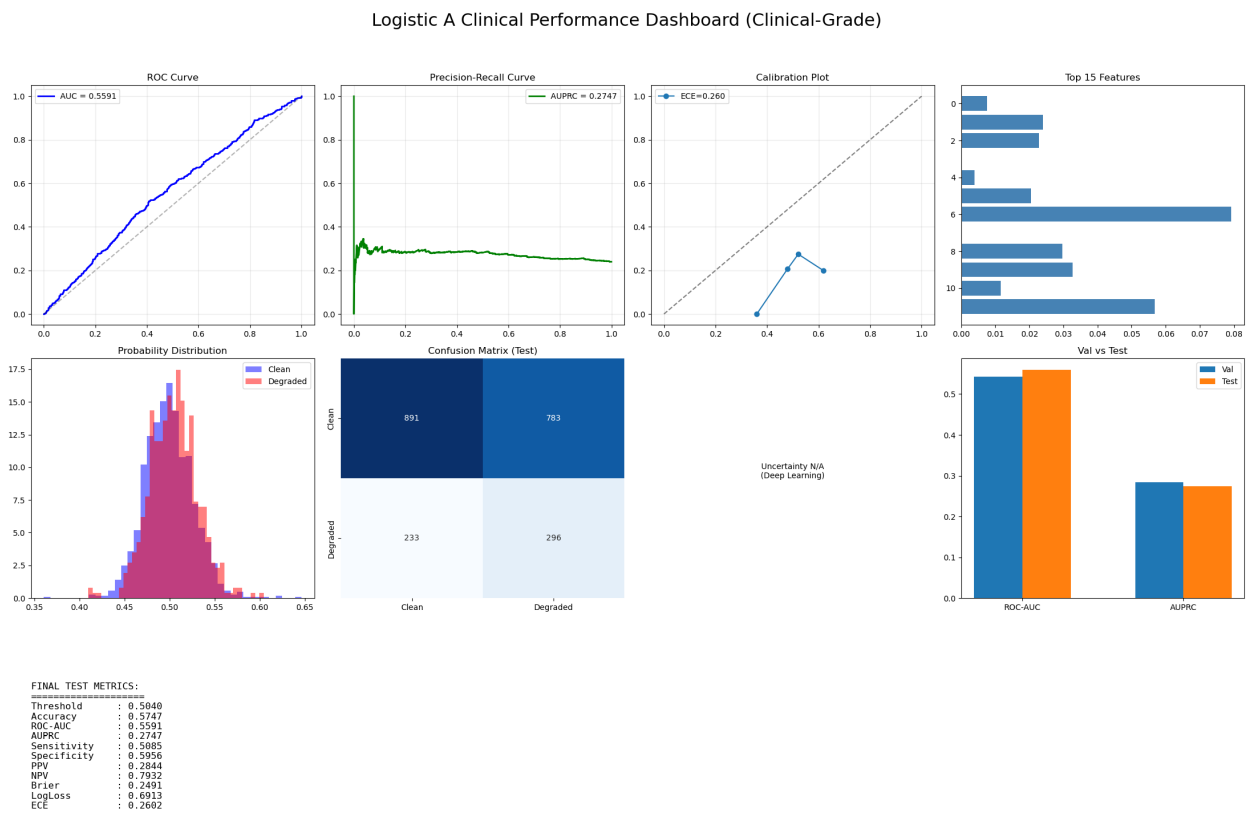


Figure 3. Performance dashboard for Model A (Logistic Regression, Basic SQI Features) showing ROC curve, precision-recall curve, calibration plot, feature importance, probability distributions, confusion matrix, uncertainty intervals, and per-class metrics.

Logistic B Clinical Performance Dashboard (Clinical-Grade)

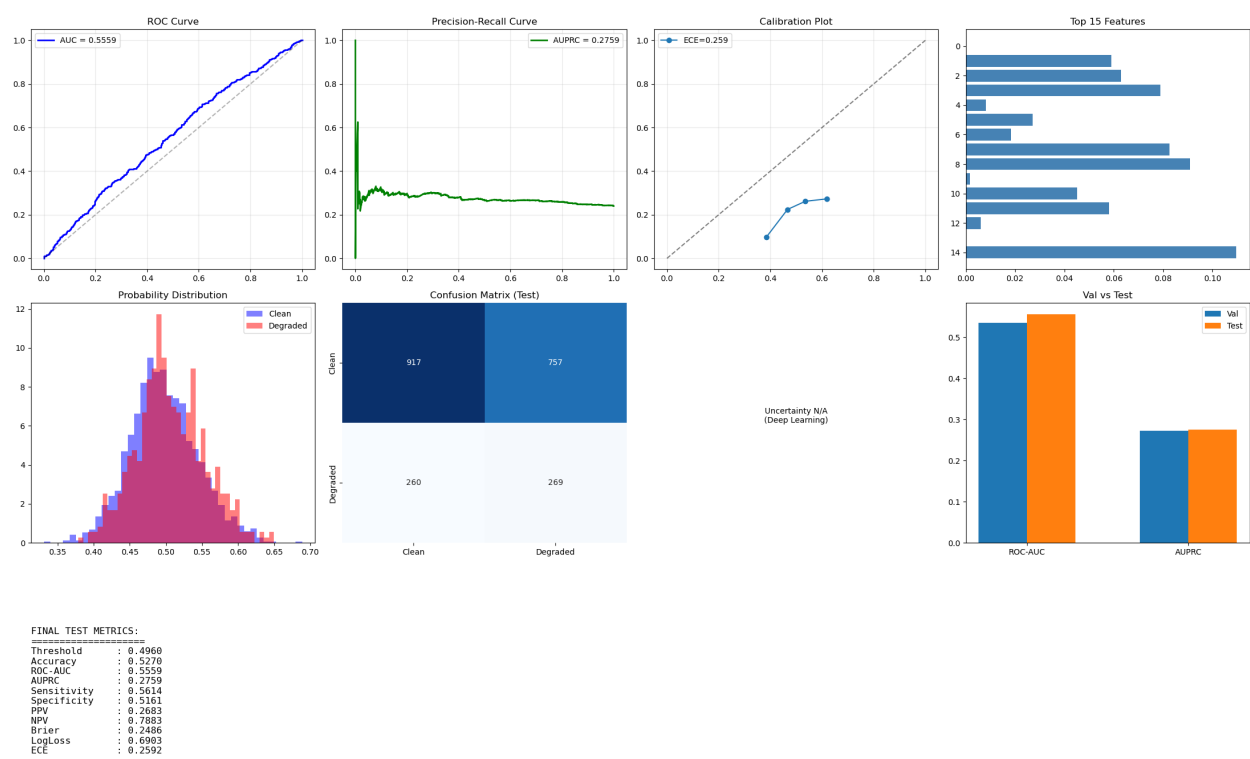


Figure 4. Performance dashboard for Model B (Logistic Regression, Enhanced SQI Features).

DL1 Clinical Performance Dashboard (Clinical-Grade)

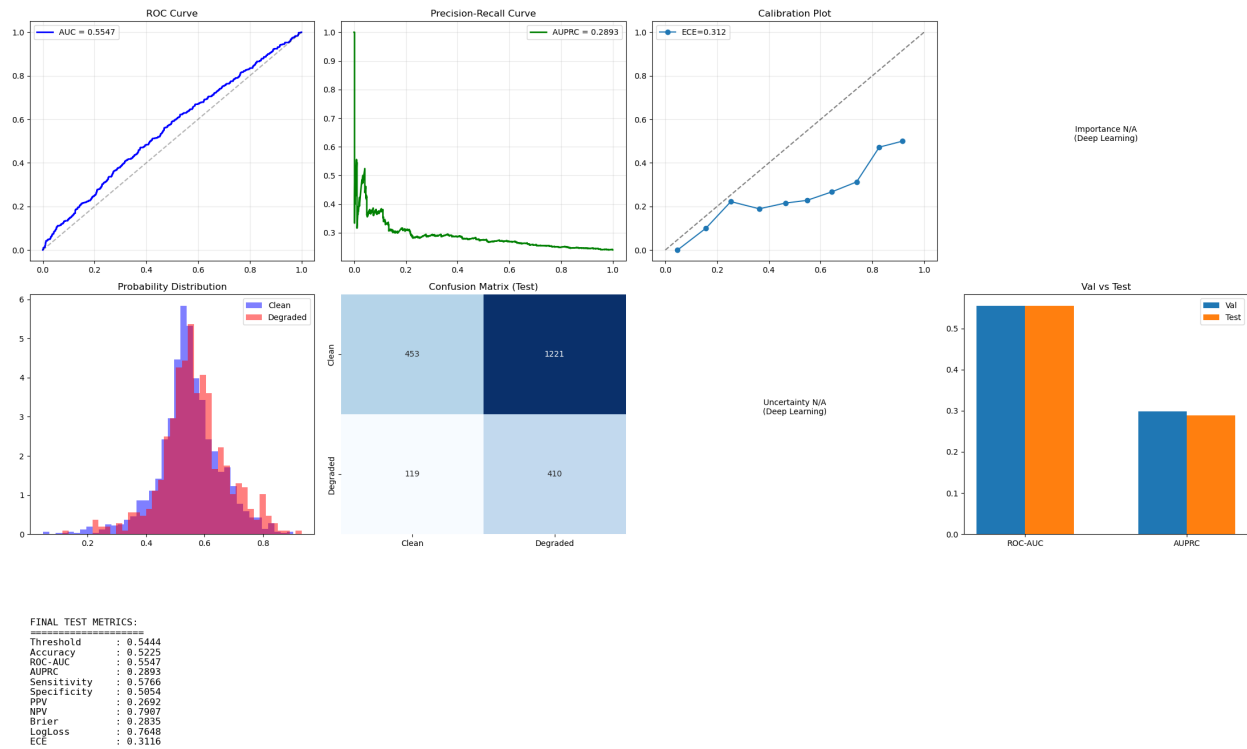


Figure 5. Performance dashboard for DL1 (Simple MLP on handcrafted features).

Deep learning advantage

Shifting focus from feature-based methods to raw-waveform, deep learning (DL2) performance improved significantly, ROC-AUC increased from 0.56 to 0.80, an absolute gain of 24 percentage points. AUPRC nearly doubled from 0.28 to 0.54. This gap demonstrates that the 1D-CNN learns features from the raw waveform that capture signal degradation patterns not accessible to traditional SQIs.

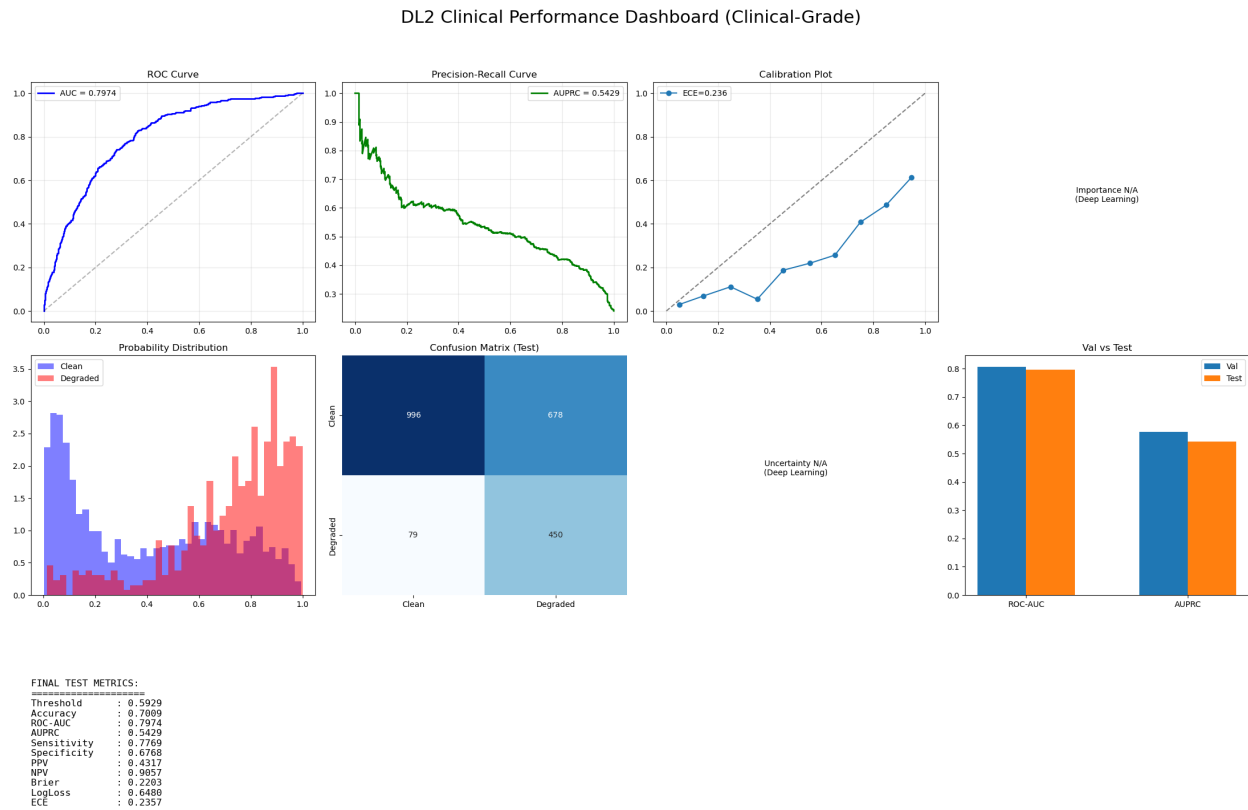


Figure 6. Performance dashboard for DL2 (Deep 1D-CNN on raw waveforms).

The final benchmark ensemble model DL3 provided an additional boost, achieving an AUC of 0.813 and an AUPRC of 0.569. While the raw discrimination improvement over DL2 was modest, DL3's primary value lies in its calibrated uncertainty estimates. The 90.2% conformal coverage (versus the target 90%) demonstrates that the prediction sets provide reliable coverage guarantees.

DL3 SOTA Ensemble Performance Dashboard (Clinical-Grade)

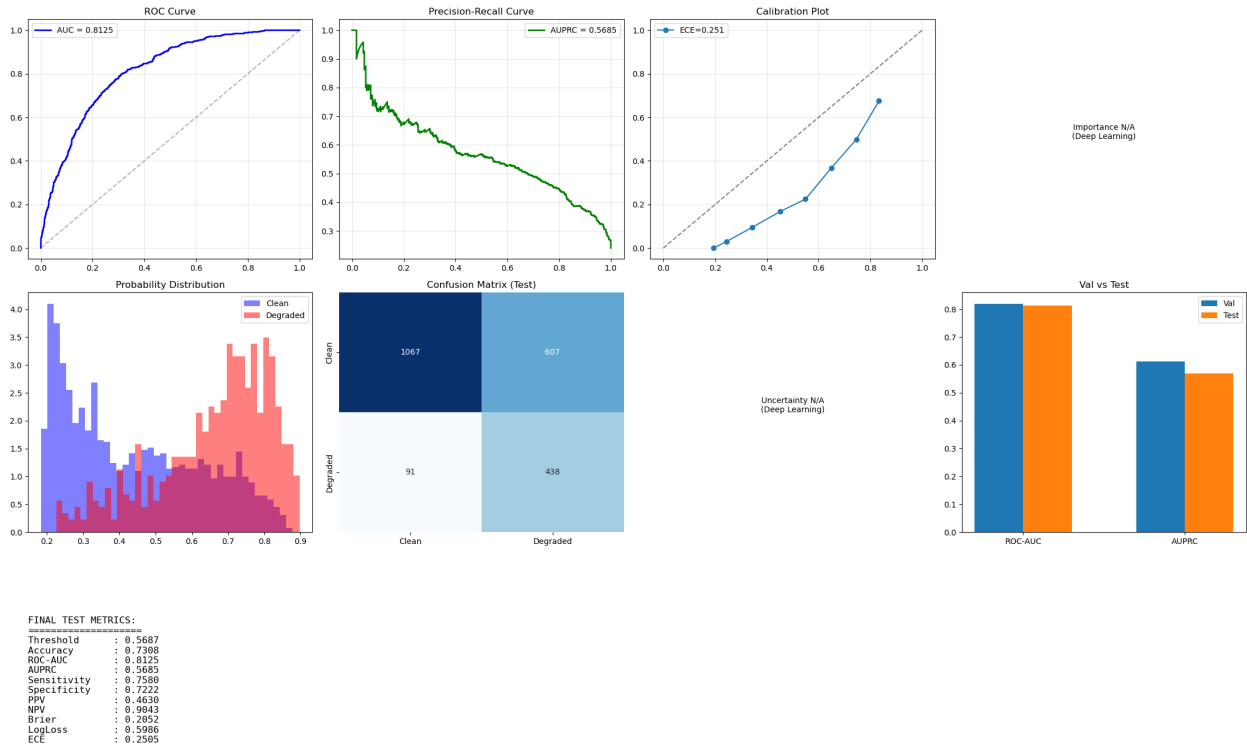


Figure 7. Performance dashboard for DL3 (CNN-Transformer Ensemble with uncertainty quantification).

Per class performance analysis

Results with the default threshold of 0.5 indicated that performance across classes was challenging for the minority class (degraded recordings). DL3 achieved a precision of 0.921, a recall of 0.637, and an F1 of 0.754 for the clean class, while the degraded class had a precision of 0.419, a recall of 0.828, and an F1 of 0.557. The model correctly identifies 82.8% of degraded recordings, but at the cost of some false positives among clean recordings. For clinical deployment, the threshold could be adjusted based on the relative costs of different error types. The cost-sensitive analysis, with a 5:1 false-negative-to-false-positive cost ratio, yielded optimal thresholds ranging from 0.25 to 0.46 for at least four of the models, with model B as an outlier at 0.05, reflecting its poor discriminatory power. The lower threshold shifts the operating point toward higher sensitivity at the expense of specificity.

Model comparison/Benchmarking

A bootstrap-based DeLong test was conducted to compare AUC pairwise against the DL3 benchmark. All comparisons were statistically significant:

Model A vs. DL3: AUC difference of -0.254 (95% CI: [-0.287, -0.221], $p < 0.001$)

Model B vs. DL3: AUC difference of -0.257 (95% CI: [-0.290, -0.224], $p < 0.001$)

DL1 vs. DL3: AUC difference of -0.258 (95% CI: [-0.290, -0.226], $p < 0.001$)

DL2 vs. DL3: AUC difference of -0.015 (95% CI: [-0.029, -0.002], $p = 0.019$)

Even the modest improvement of DL3 over DL2 was statistically significant, despite the practical difference being small.

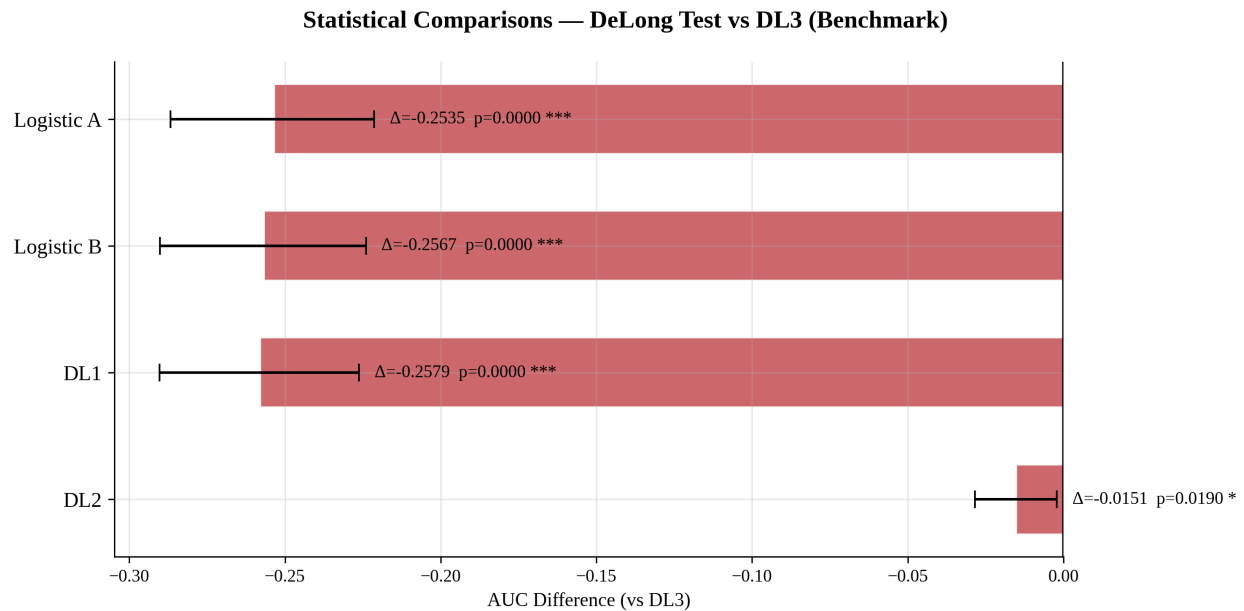


Figure 8. DeLong test results comparing each model's AUC against the DL3 benchmark, with 95% confidence intervals and significance levels.

Uncertainty estimates

The DL3 ensemble provides decomposed uncertainty estimates. Mean epistemic uncertainty (model variability) was 0.062 ± 0.038 , while mean aleatoric uncertainty (predicted data noise) was 1.001 ± 0.044 . The relatively low epistemic uncertainty suggests that the ensemble members have converged on similar representations, while the higher aleatoric uncertainty indicates inherent ambiguity in many recordings.

The relationship between estimated uncertainty and predicted accuracy was evaluated by examining model performance at different confidence thresholds, and the results indicated that accuracy increases as more samples are included. The aleatoric uncertainty may not fully capture the data-level noise characteristics. Despite this limitation, the conformal prediction framework provides a more practically useful guarantee: 90.2% empirical coverage at the 90% target level, meaning prediction sets reliably contain the accurate label regardless of the uncertainty head's calibration.

Discussion

The goal of this project was to assess whether literature-based signal quality measures can learn patterns that distinguish between a clean and a degraded signal, regardless of the diagnostic class and algorithm. Beyond that, provide insights into how raw signals compare with these handcrafted signal quality measures. The project also quantified model uncertainty with the benchmark model.

Deep learning's superior performance

Findings demonstrated a performance gap between the handcrafted signal quality measures and the raw waveform models. The performance gap observed calls for a thorough look at what deep learning models capture that handcrafted SQIs miss. There could be a number of reasons for this observation. First, traditional SQIs are designed to be robust to normal physiological variation, which may inadvertently reduce their sensitivity to specific noise patterns. In the paper by Campero and colleagues in 2023, SQI metrics such as skewness and signal-to-noise ratio (SNR) typically perform well with noise-free data which explains our observations, making our model results susceptible to both noise and actual morphological variations (Satija et al., 2018), such as bundle branch blocks. Rahman and colleagues also reported inconsistent results across datasets for metrics such as kurtosis, skewness, and power spectrum distribution (Rahman et al., 2022). Another reason is the ability of CNNs to learn the discriminative features specifically needed for the quality classification objective. In a review by Jaya Prakash et al. (2025), they showed that deep learning architectures and CNNs learn features required for the classification task and eliminate the need for manual feature engineering, further underscoring deep learning's superior performance. Besides this, multi-scale convolutional architecture can capture noise patterns at different temporal scales simultaneously. With varying kernel sizes in convolution layers, different types of noise and artifacts can be captured since different noise types exhibit different frequencies. Furthermore a convolution operation have been shown to use a dilation factor to increase receptive fields to capture the temporal structure of ECG noise at multiple scales (Mvuh et al., 2024; Zhou & Fang, 2025; Zhu et al., 2024). Burst noise for instance, may last milliseconds while baseline drift unfolds over seconds. The parallel kernel sizes in DL3's stem explicitly model this multi-scale structure for burst noise and baseline drifts. Finally, the transformer's self-attention facilitates the identification of relationships between different parts of the recording (Che et al., 2021). When noise appears intermittently, the model can learn to compare clean and noisy segments within the same recording, a computation that aggregated SQI statistics cannot perform.

Interpretable baseline

Interpretable models still have a place in signal quality detection, despite their relatively poor performance. They establish baselines against which deep learning improvements can be measured. This has been demonstrated in previous research by Zhang et al. (2021), who found that deep learning models outperformed traditional machine learning models, with the machine learning models serving as a baseline. Additionally, examination of feature coefficients indicates which signal characteristics are associated with quality degradation, providing scientific insight even if classification accuracy is limited. This provides the basis and scientific understanding of what black-box models such as deep learning may be learning, beyond the interpretability of the handcrafted features (Jaya Prakash et al., 2025). Moreover, in resource-limited settings where deep learning models may be impractical, the feature-based models offer lightweight alternatives. Simpler models still have great value, especially in memory, power, and cost-constrained environments, making lighter-weight models more relevant (Ansari et al., 2023; Hizem et al., 2025; Tan et al., 2022). It should be noted that while the original experimental design envisioned fully Bayesian logistic regression with horseshoe priors for principled uncertainty quantification, convergence difficulties during development led to the adoption of L1-regularized logistic regression as a practical alternative. This challenge has been discussed previously by Piironen and Vehtari (2017). The authors discuss convergence difficulties and propose careful hyperparameter tuning. The Bayesian formulation remains a promising direction for future work, as it would provide posterior credible intervals over model coefficients and per-prediction uncertainty without requiring ensemble methods.

Clinical trust

The expected calibration error (ECE) values (0.24-0.31) showed that all models exhibit some degree of miscalibration, i.e., a mismatch between predicted probabilities and empirical frequencies. DL2 had the best raw ECE (0.236), while the temperature-scaled DL3 achieved 0.251. These values are reasonable but suggest room for improvement through additional calibration techniques or training with calibration-aware losses.

The conformal prediction framework is valuable for clinical deployment. The guaranteed coverage property means that clinicians can trust that, on average, 90% of the time, the true quality label will be contained in the prediction set. When the model produces a singleton prediction set, it is confident, but when it produces an ambiguous set, it explicitly warrants human review. In clinical settings, the model should be based on an individual level, hence it is non-negotiable to present a score without an associated confidence interval (Vazquez & Facelli, 2022), and this also supports risk-based decision-making.

Recommendation for deployment

Based on the study results, the following recommendations for a deployment strategy for ECG signal quality assessment are proposed

1. At acquisition time, DL3 ensemble (or DL2 if computational resources are limited) should be deployed to provide immediate feedback on signal quality. Recordings flagged as degraded with high confidence can trigger automated reacquisition prompts. This serves as a quality control measure for users, whether remote or the wearer of a wearable device monitor, to ensure optimal use and to allow inexperienced users to use the device with confidence (Neri et al., 2024; Zhou et al., 2021).
2. For uncertain predictions, recordings whose prediction set contains both classes should be flagged for human review, especially in settings with limited medical professionals. In these settings, medical professionals must review multiple data points to make critical healthcare decisions, which can be time-consuming (Liu et al., 2023). This selective routing ensures expert attention is focused where it is most needed.
3. Ultimately, downstream algorithms can use quality scores to weight the confidence of automated diagnoses. A high-quality ECG supporting an abnormal finding carries more weight than a degraded recording with the same algorithmic output.

Threshold selection

The choice between sensitivity and specificity in clinical decision-making depends on the cost of a false-positive relative to a false-negative. Cutoffs are usually based on statistical analyses that find an acceptable balance between the two choices (Armstrong & Metlay, 2020), as the consequences vary with the Youden index and provide limited value for determining clinical consequences.(Mah et al., 2024). Hence, the optimal classification threshold should depend on the clinical context and the relative costs of different error types. For instance, a high-sensitivity setting (threshold ~ 0.38) should be employed in screening contexts where missing a degraded recording could lead to diagnostic errors. A more balanced (threshold ~ 0.57 , Youden optimal) is appropriate for general clinical use where both error types are equally costly. Then, a high-specificity setting (threshold ~ 0.70) is recommended when reacquisition is costly or impractical (e.g., ambulatory monitoring), and only recordings that are clearly degraded should be flagged.

Clinical workflow integration

Signal quality assessment should integrate seamlessly with existing ECG management infrastructure. One key recommendation from this study is to employ three model outputs during deployment: 1. both a binary flag (clean/degraded), 2. a continuous quality score (probability of degradation), and 3. an uncertainty indicator. Automated systems may rely on the binary flag for

simple routing decisions. As previously noted, SQI metrics remain a valuable tool for quality control prior to signal analyses and, when integrated into automated workflows, can help reduce medical professionals' workload (Kuetché et al., 2023; van der Bijl et al., 2022). At the same time, quality-aware algorithms can incorporate the continuous score as a weighting factor, and clinicians can use the uncertainty indicator to prioritize manual review. The goal of automated systems is not to eliminate the need for human experts but rather to augment the workload, as shown by previous research that the inclusion of an uncertainty indicator significantly improved the accuracy of diagnosis, as cases with a higher uncertainty score were referred to human experts (Vranken et al., 2021)

Study limitations

The current study is not without limitations. First, the PTB-XL dataset was collected at a single institution using specific equipment; hence, generalization to other devices, patient populations, and clinical settings requires external validation and must be done with caution. This limitation was noted by Phan et al. (2025), who proposed a cross-dataset experiment. Second, our binary quality labels aggregated multiple noise types (baseline drift, burst noise, electrode problems) into a single class. A multi-label approach distinguishing specific artifact types could provide more actionable information (Satija et al., 2018) since different noise types exhibit distinct characteristics (Zhang et al., 2025). Third, while we attempted fully Bayesian neural network approaches with horseshoe priors during development, convergence difficulties led us to adopt the simpler L1-regularized logistic regression for the interpretable models. A fully Bayesian deep learning approach with proper posterior uncertainty quantification remains a promising direction for future work. Also, our models operate on complete 10-second recordings; higher-resolution temporal localization of noise segments could improve interpretability and enable targeted artifact correction (Satija et al., 2018). Finally, the DL3 ensemble's learned uncertainty decomposition showed limited correlation with prediction accuracy, with even the most confident 25% of predictions achieving only 52.3% accuracy. This suggests that while the conformal prediction framework provides reliable coverage guarantees, the model's internal uncertainty estimates require further refinement before they can be used as standalone confidence indicators. One reason could be that DL3 cannot completely separate epistemic and aleatoric uncertainties, which is a challenging task (Vranken et al., 2021).

Future Direction

Several promising directions emerge from this work. Multi-task learning can jointly predict quality labels and diagnostic classes by leveraging shared representations (see Geng et al., 2023). Transfer learning from diagnostic pre-trained models could improve quality assessment, especially when labeled quality data is scarce. Transfer learning reduces computational

complexity while achieving comparable performance to both deep and shallow networks (Weimann & Conrad, 2021; Zhou et al., 2021). Explainability techniques like GradCAM could visualize which waveform regions drive quality predictions, increasing clinical trust. Real-time deployment on edge devices (ECG machines, wearables) would enable immediate quality feedback during acquisition. Finally, prospective clinical validation studies are essential before deployment, measuring not only classification accuracy but also the downstream impact on diagnostic outcomes.

Conclusion

The goal of this project was to investigate ECG signal quality classification using five models spanning interpretable logistic regression methods and state-of-the-art deep learning. Our key findings are as follows:

It was observed that handcrafted signal quality indices, despite their foundation in the ECG signal processing literature, exhibit a ceiling effect of approximately 56% AUC across classification algorithms. This ceiling cannot be broken by adding more features, using more sophisticated classifiers, or employing non-linear transformations, showing that traditional SQIs do not capture the information required for accurate quality assessment.

Furthermore, deep learning models operating directly on raw 12-lead waveform data substantially outperform feature-based approaches, achieving ROC-AUCs of up to 0.813. The learned representations capture degradation patterns that feature engineering cannot encode. For practical ECG quality assessment, end-to-end deep learning is the methodology of choice.

Third, uncertainty quantification through ensemble methods and conformal prediction provides valuable insight for clinical deployment. The DL3 model's 90.2% conformal coverage demonstrates that distribution-free prediction sets can provide reliable coverage guarantees, enabling trustworthy decision support.

As ECG-based clinical decision support systems become more prevalent, automated signal quality assessment will serve as an essential gatekeeper, ensuring that only reliable data informs critical clinical decisions. This work provides both the methodological framework and empirical evidence to guide the development and deployment of such systems.

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